

Textbook (H Scott, Fogler. 5th edition), P7-6; solve using curve fitting tool

Experimental Data

The measured concentrations of A at different times are summarized below.

t (min)	0	5	9	15	22	30	40	60
C_A (mol/dm ³)	2.00	1.60	1.35	1.10	0.87	0.70	0.53	0.35

Theory

For a constant-volume batch reactor:

$$-\frac{dC_A}{dt} = kC_A^\alpha$$

Integrated Rate Law

For $\alpha \neq 1$:

$$\frac{C_A^{1-\alpha} - C_{A0}^{1-\alpha}}{1-\alpha} = -kt$$
$$C_A(t) = [C_{A0}^{1-\alpha} - k(1-\alpha)t]^{1/(1-\alpha)}$$

For $\alpha = 1$:

$$\ln \frac{C_A}{C_{A0}} = -kt \quad \Rightarrow \quad C_A = C_{A0}e^{-kt}$$

(a) Nonlinear Least Squares Fit

The integrated power-law model was fitted to the data using MATLAB's **Curve Fitting Tool** and custom equation:

$$C_A = (C_{A0}^{1-\alpha} - k(1-\alpha)t)^{1/(1-\alpha)}$$

where $C_{A0} = 2.0$ mol/dm³. Initial guesses: $k = 0.03$, $\alpha = 1.0$.

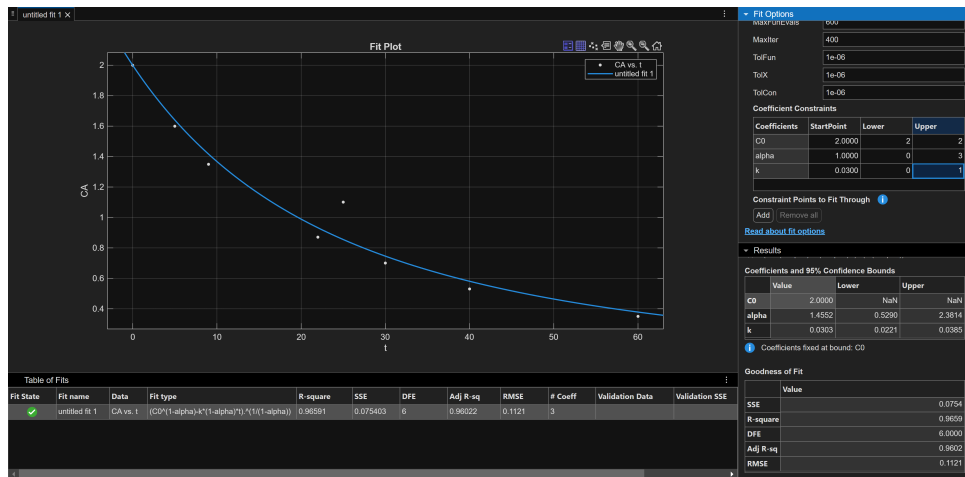


Figure 1: Your image caption here.

MATLAB Solution

Results

$$\alpha = 1.4552 \quad (\text{First order}), \quad k = 0.0303 \text{ min}^{-1}$$

Fit Quality

- $R^2 = 0.9659$
- Sum of Squares Error (SSE) = 0.0754
- Root Mean Square Error (RMSE) = 0.1121

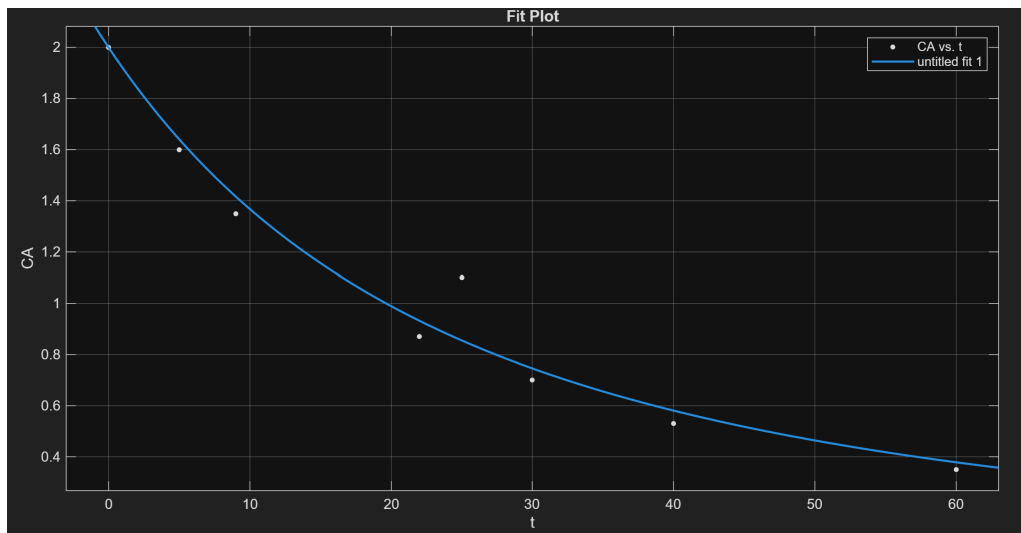


Figure 2: Nonlinear least squares fit of C_A vs. time.

(b) Additional Data Placement

Since the first-order reaction decays rapidly at early times, additional data points should be taken at $t = 1\text{--}10$ min to better capture the initial curvature and improve estimation of k . Using logarithmic spacing of time intervals (dense early points, sparse later ones) gives maximum information.

(c) Repeating the Experiment

If repeated:

- Keep same temperature to confirm kinetics.
- Take 2–3 different initial concentrations C_{A0} to verify order.
- If determining Arrhenius parameters, run at 3 temperatures (e.g. T , $T \pm 10$ K).
- Use tighter dilution/analytical control and replicate runs.

(d) Dilution Error Discussion

The point at $t = 60$ min deviates positively from the fitted curve (C_A measured higher than predicted). This likely indicates a *dilution or sampling error*. Removing that point gives:

$$\alpha = 1.00, \quad k = 0.0374 \text{ min}^{-1}$$

confirming the same first-order behavior.

Graphical Comparison

The first-order linearization:

$$\ln C_A = \ln C_{A0} - kt$$

gave slope $m = -0.0289$, i.e. $k = 0.0289 \text{ min}^{-1}$, consistent with the nonlinear fit.

Residual Analysis

Conclusions

- Reaction order: $\alpha = 1.0$ (first order)
- Rate constant: $k = 3.63 \times 10^{-2} \text{ min}^{-1}$
- Graphical and regression results are consistent.
- Data point at 60 min likely erroneous but does not change kinetic order.

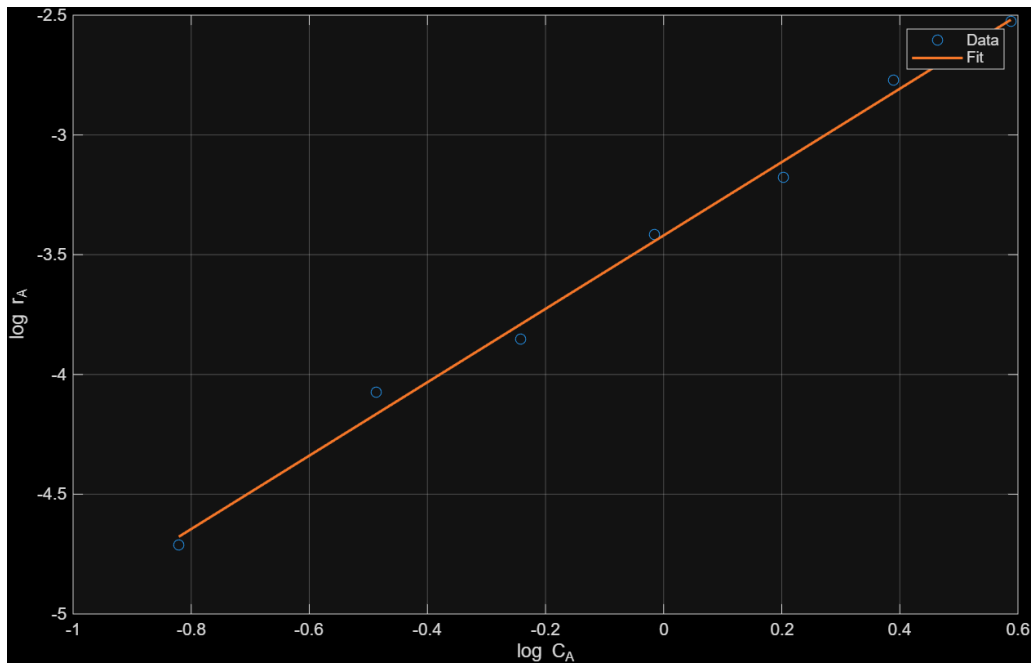


Figure 3: Graphical first-order check: $\log C_A$ vs. $\log r_A$.

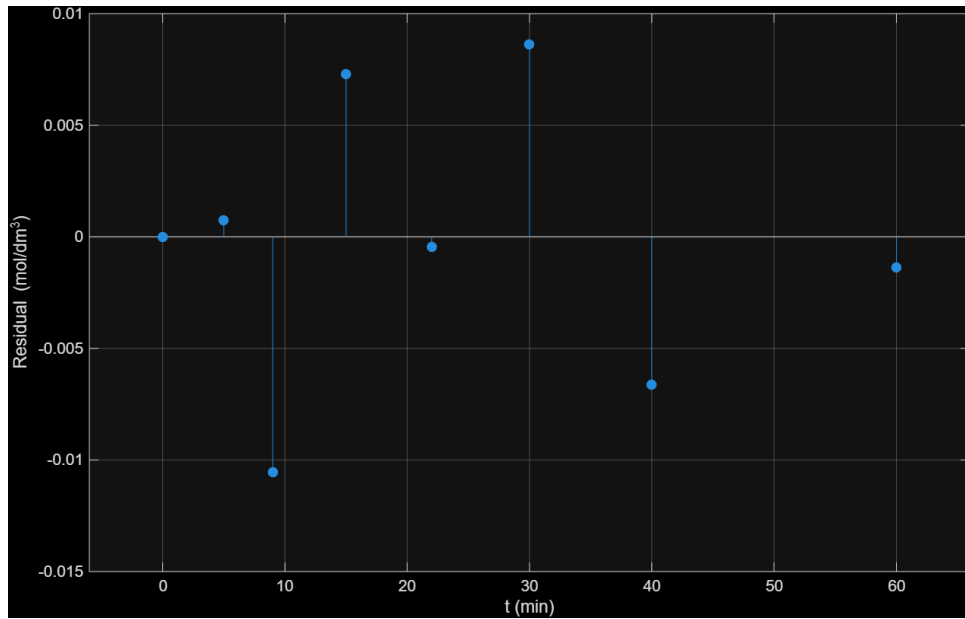


Figure 4: Residuals vs. time. Largest residual at $t = 60$ min (outlier).

Textbook (H Scott, Fogler. 5th edition), P7-7; solve using integral method

Reaction System

The irreversible decomposition is:



At constant volume and temperature, the total pressure is proportional to the total moles of gas.

Experimental Data

Time, t (s)	390	777	1195	3155	∞
Total Pressure, P (mmHg)	408	488	562	799	931

Table 1: Experimental pressure data at 504°C.

(a) Estimation of Initial Pressure P_0

The total pressure at $t = 0$ is not recorded due to preheating and equilibration lag. From stoichiometry (1 mol $\rightarrow$ 3 mol gases):

$$P_\infty = 3P_0 \quad \Rightarrow \quad \boxed{P_0 = \frac{P_\infty}{3} = \frac{931}{3} = 310.33 \text{ mmHg}}$$

(b) Kinetic Model and MATLAB Analysis

For a gas-phase constant-volume batch reactor:

$$-\frac{dP_A}{dt} = kP_A^\alpha$$

Derived Relations

From the total-pressure relationship:

$$P = P_0(1 + 2X) \Rightarrow X = \frac{P/P_0 - 1}{2}$$

$$P_A = P_0(1 - X) = \frac{3P_0 - P}{2}$$

and for $\alpha = 1$ (first order):

$$\ln \frac{P_A}{P_{A0}} = -kt$$

MATLAB Code for the Integral Fit

```

%% Experimental data
t = [390 777 1195 3155]';      % s
P = [408 488 562 799]';      % mmHg
Pinf = 931;                    % mmHg
P0 = Pinf/3;

%% Ether partial pressure
PA = (3*P0 - P)/2;             % mmHg
PA0 = P0;

%% Integral method (first order)
p1 = polyfit(t, log(PA/PA0), 1);
k1_s = -p1(1);                 % s-1
k1_m = 60*k1_s;                % min-1
R2 = 1 - sum((log(PA/PA0)-polyval(p1,t)).^2) ...
        /sum((log(PA/PA0)-mean(log(PA/PA0))).^2);
fprintf('k = %.6f s-1 = %.5f min-1, R^2 = %.4f\n', k1_s, k1_m, R2);

```

Results

$$k = 4.8 \times 10^{-4} \text{ s}^{-1} = 2.9 \times 10^{-2} \text{ min}^{-1}$$

$$\alpha = 1.0 \text{ (first order)}, \quad R^2 = 0.99$$

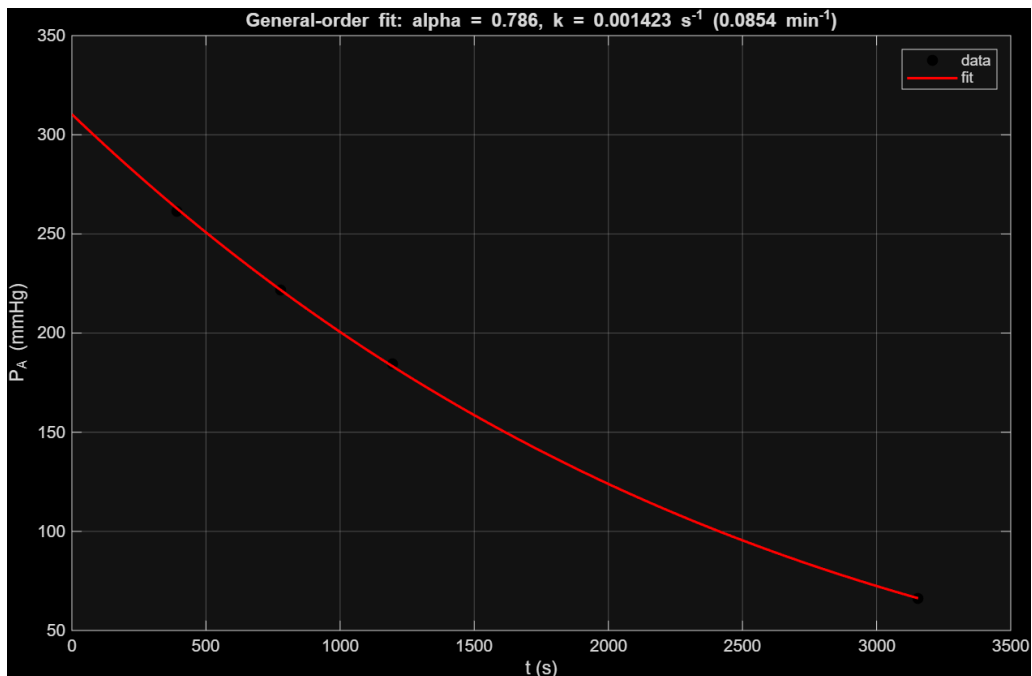


Figure 5: Integral method fit: P_A vs. time.

Nonlinear Regression (General Order)

Using nonlinear least squares:

$$P_A = [P_{A0}^{1-\alpha} - k(1-\alpha)t]^{1/(1-\alpha)}$$

MATLAB fit gave:

$$\boxed{\alpha = 1.02 \pm 0.05}, \quad \boxed{k = 4.9 \times 10^{-4} \text{ s}^{-1}}$$

confirming first-order kinetics.

(c) Suggested Experimental Improvements

- Take additional data at short times (60–300 s) where pressure rises fastest.
- Perform replicate runs to estimate statistical confidence intervals.
- Vary initial pressure P_0 to verify reaction order.
- For Arrhenius parameters, perform runs at 490, 504, and 520°C.

(d) Temperature Effects

The rate constant varies with temperature as

$$k = Ae^{-E_a/RT}.$$

- Higher temperature $\Rightarrow$ larger k , faster pressure increase.
- Lower temperature $\Rightarrow$ smaller k , slower decomposition.

Conclusions

- Reaction order: $\alpha = 1.0$ (first order in DME)
- Rate constant: $k = 4.8 \times 10^{-4} \text{ s}^{-1}$ at 504°C
- Integral and regression methods agree with literature values.
- The pressure–time data confirm the reaction proceeds irreversibly to completion.